

CODE:

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from PIL import Image
import numpy as np
from sklearn.cluster import MiniBatchKMeans
import matplotlib.pyplot as plt
import math

image_path = "C:/Users/dhivy/OneDrive/Desktop/sem- 6/DIP/Input (image).jpg"

img = Image.open(image_path).convert("RGB")
arr = np.asarray(img, dtype=np.float32) / 255.0
h, w, c = arr.shape
X = arr.reshape(-1, c)

N = X.shape[0]
sample_size = min(10000, N)
idx = np.random.choice(N, sample_size, replace=False)
X_sample = X[idx]

K_values = [4, 8, 16, 32, 64]
rates = []
mses = []
psnrs = []

for K in K_values:
    kmeans = MiniBatchKMeans(n_clusters=K, batch_size=2048, max_iter=50, n_init=1)
    kmeans.fit(X_sample)
    labels = kmeans.predict(X)
    centers = kmeans.cluster_centers_
    Xq = centers[labels]
    quant_arr = Xq.reshape(h, w, c)
    diff = arr - quant_arr
    mse = np.mean(diff ** 2)
    mses.append(mse)
    if mse == 0:
        psnr = float("inf")
    else:
        psnr = 10 * math.log10(1.0 / mse)
    psnrs.append(psnr)
    counts = np.bincount(labels, minlength=K).astype(np.float64)
    p = counts / counts.sum()
    p = p[p > 0]
    entropy = -np.sum(p * np.log2(p))
    rates.append(entropy)
    if K in [4, 16, 64]:
        plt.figure()
        plt.imshow(quant_arr)
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plt.title(f"K = {K}, PSNR = {psnr:.2f} dB, R = {entropy:.3f} bits/pixel")
plt.axis("off")

rates = np.array(rates)
mses = np.array(mses)
psnrs = np.array(psnrs)

plt.figure()
plt.plot(rates, psnrs, marker="o")
for i, K in enumerate(K_values):
    plt.text(rates[i], psnrs[i], f"K={K}")
plt.xlabel("Rate (bits/pixel)")
plt.ylabel("PSNR (dB)")
plt.title("Fast K-means Rate–Distortion Curve")
plt.grid(True)
plt.show()

print("K\tRate(bits/pixel)\tMSE\t\tPSNR(dB)")
for K, R, mse, psnr in zip(K_values, rates, mses, psnrs):
    print(f"{K}\t{R:.4f}\t\t{mse:.6f}\t{psnr:.2f}")

```